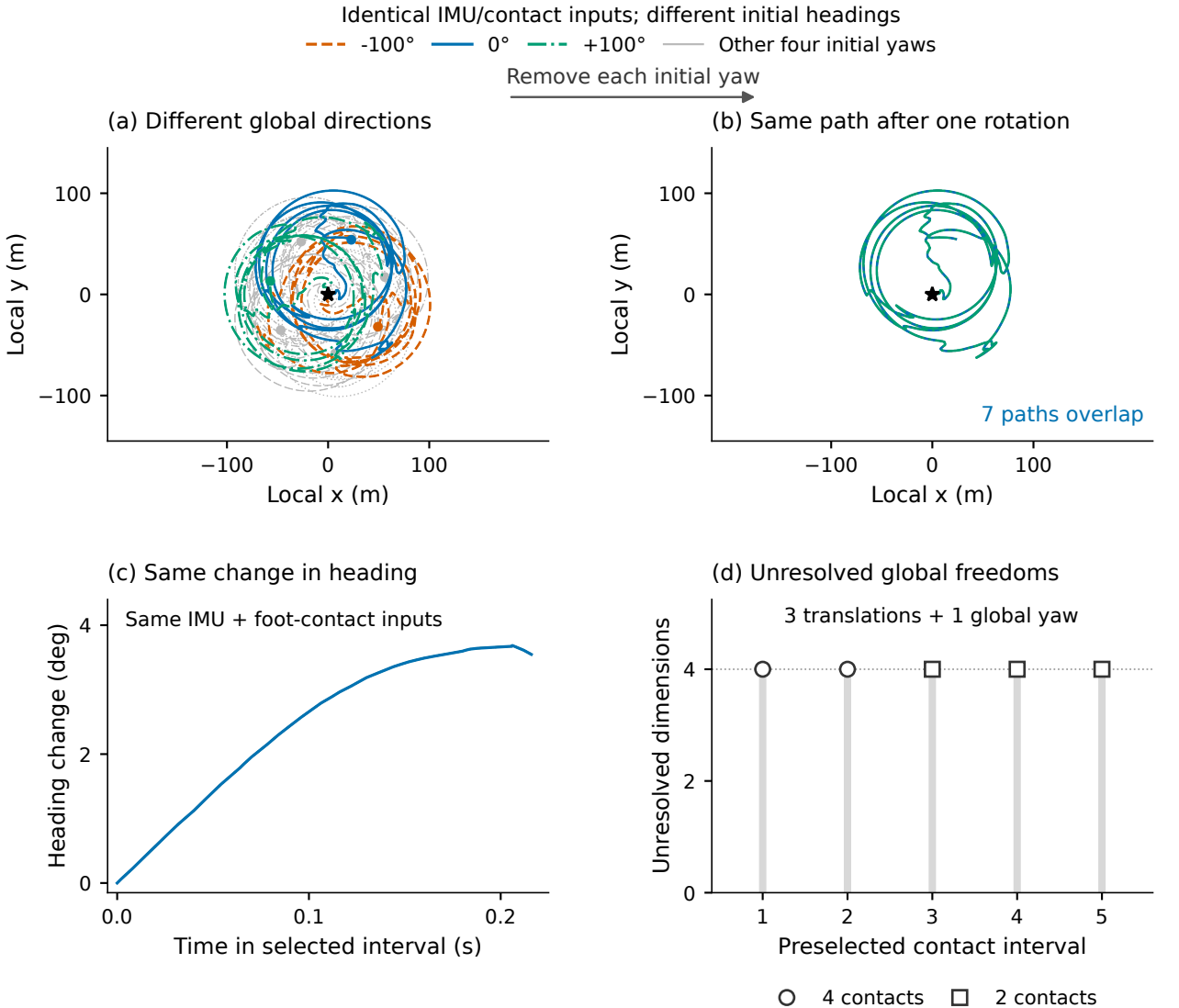

Identical IMU/contact inputs; different initial headings
-100°
0°
+100°
Other four initial yaws
Remove each initial yaw
(a) Different global directions
Local y (m)
Local x (m)
100
0
−100
(b) Same path after one rotation
7 paths overlap
(c) Same change in heading
Same IMU + foot-contact inputs
Heading change (deg)
Time in selected interval (s)
0.0
0.1
0.2
(d) Unresolved global freedoms
3 translations + 1 global yaw
Unresolved dimensions
Preselected contact interval
1
2
3
4
5
4 contacts
2 contacts